Compressed OOPs

32-bit references on a 64-bit JVM

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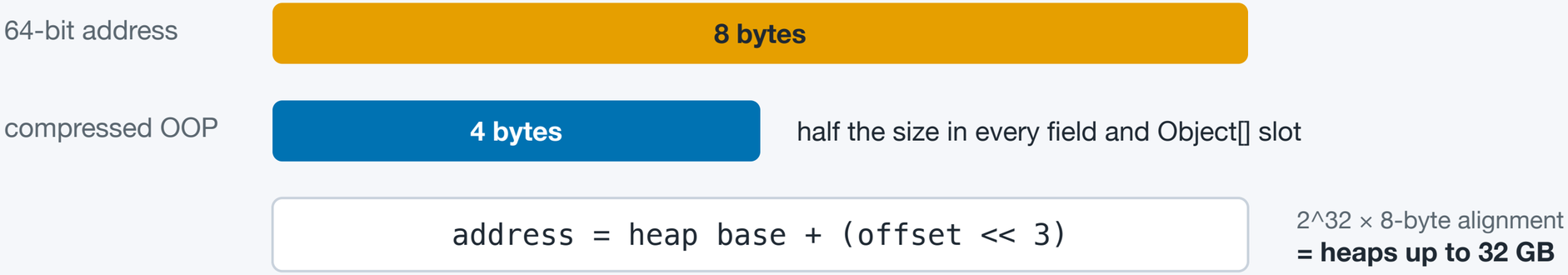
What is a compressed OOP?

A 4-byte offset that HotSpot stores instead of an 8-byte reference.

- OOP = ordinary object pointer: a reference to a Java object.
- The offset is scaled by the 8-byte object alignment.
- On by default for heaps under 32 GB (`-XX:+UseCompressedOops`).

Half the size per reference

A reference field: uncompressed vs compressed



- Every reference field and `Object[]` slot shrinks from 8 to 4 bytes.
- Example: an `ArrayList` of 1M elements needs a 4 MB array, not 8 MB.

References

- Java HotSpot VM Performance Enhancements (Java SE 25): Compressed Ordinary Object Pointer
- The `java` command (Java SE 25): `-XX:+UseCompressedOops`